# Imbalance Price Forecast

FARMERS · Turku Energia

*Level 1 Evaluation Report*

## 1 Introduction

Our Long-Short Term Gradient Boosting Model (LSTGBM) splits each 10-hour forecast between two LightGBM models. A specialist handles the first eight quarter-hours and adds recent Fingrid measurements to the shared inputs. A second model handles quarters 9–40 using the shared inputs alone. Each model estimates how far the imbalance price could move from spot under up- or down-regulation, then combines those scenarios using the predicted odds of each direction.

## 2 How LSTGBM Works

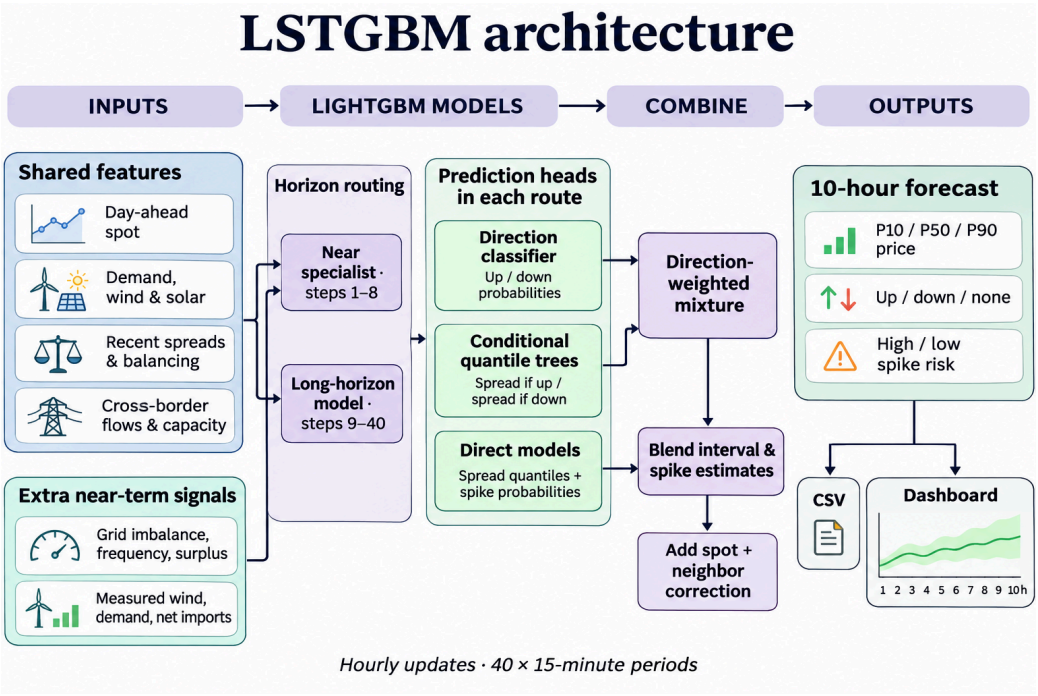



*Figure 1. LSTGBM turns spot prices and grid signals into a 10-hour forecast for the CSV and dashboard.*

**Inputs and timing.** Both models use the published day-ahead price, demand and renewable forecasts, recent balancing conditions, and cross-border flows and capacity. The first two hours also use recent Fingrid measurements, such as grid imbalance and measured wind. The longer-horizon model uses the shared inputs alone.

**What the models predict.** A classifier estimates the odds of up- and down-regulation. Separate LightGBM trees learn how far the imbalance price tends to

move away from spot in each case. Direct models independently estimate the spread's quantiles and the chance of a large move in either direction.

**From predictions to forecast.** The model mixes the two spread distributions using the direction odds. It takes the median from the mixture and blends the interval and spike estimates with the direct estimates. It adds the published spot price and applies a small correction from neighboring spot prices to the price quantiles. The CSV and dashboard show the median, the uncertainty range, and the odds of a spike or each balancing direction. The probability of no dominant direction is fixed at 0.2%.

## 3 Why we Chose this Architecture

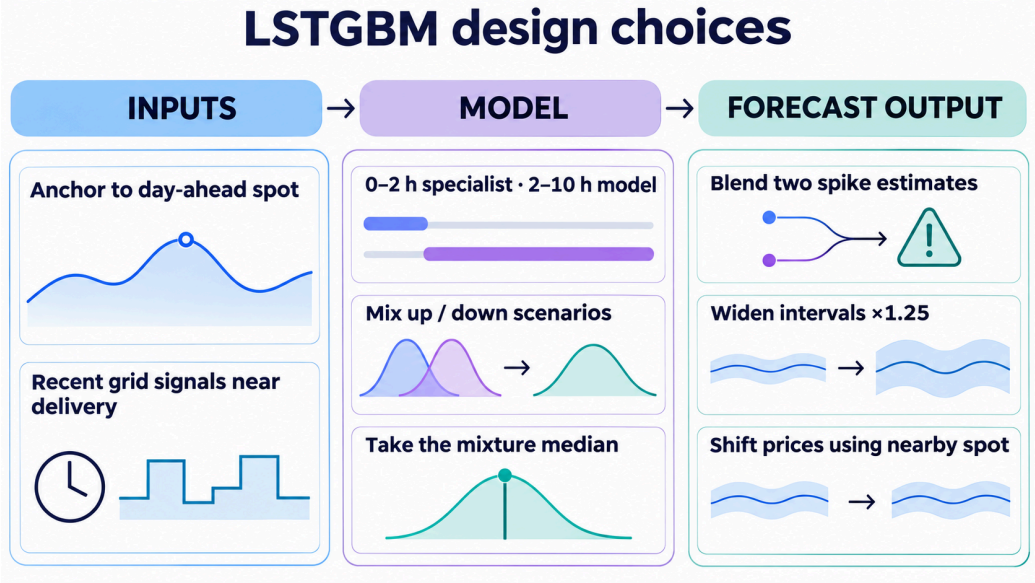



*Figure 2. LSTGBM design choices by stage.*

**Start with the known price.** Finland's imbalance price depends on the dominant balancing direction and falls back to a price based on day-ahead spot when no direction dominates. We therefore use each quarter's published spot price as an anchor and learn the spread separately for up- and down-regulation. We expected this to preserve the risk of a large move in either direction when the median alone looks unremarkable.

**Choose features that describe the Finnish system.** Forecast demand minus wind and solar gives a rough picture of the demand that other generation or imports must cover. Recent spreads and reserve prices indicate how costly balancing has been, while cross-border flows and capacity give context on whether neighboring

systems can help. Together, these signals help estimate the Finnish settlement price Turku Energia faces when it has a residual imbalance.

**Split the forecast where recent readings are most useful.** A shortage measured now can still matter over the next few quarters, but its relevance can fade as demand, generation, and power flows change. We hypothesized that a first-two-hour specialist would benefit from measured imbalance, frequency, wind, and other recent signals. For hours 3–10, we use shared inputs such as demand and generation forecasts that describe the delivery quarter itself.

**Blend two estimates of spike risk.** The direct classifiers learn which signals precede a spread beyond ±€50/MWh. The scenario model estimates the same risk from the price spread under each balancing direction and the odds of that direction. We blend them so the alert reflects both sources of information.

**Take the median after mixing.** An average of the two scenario medians can land at a spread that is unusual in either scenario. Taking the median of the combined distribution lets the point forecast reflect both the likely size of a price move and the odds of each direction.

**Account for the shape of the spot curve.** Two quarters with the same spot price can sit on different price ramps. A small correction uses published spot prices 15 and 30 minutes on either side to shift all three price quantiles together. It preserves interval width and leaves the spike probabilities unchanged.

**Widen intervals that were too narrow.** We widen the estimated intervals by 25% before blending them. On September validation, the final P10–P90 range contained 81.3% of actual prices, close to its 80% target.

## 4 Data Sources Driving the Forecast

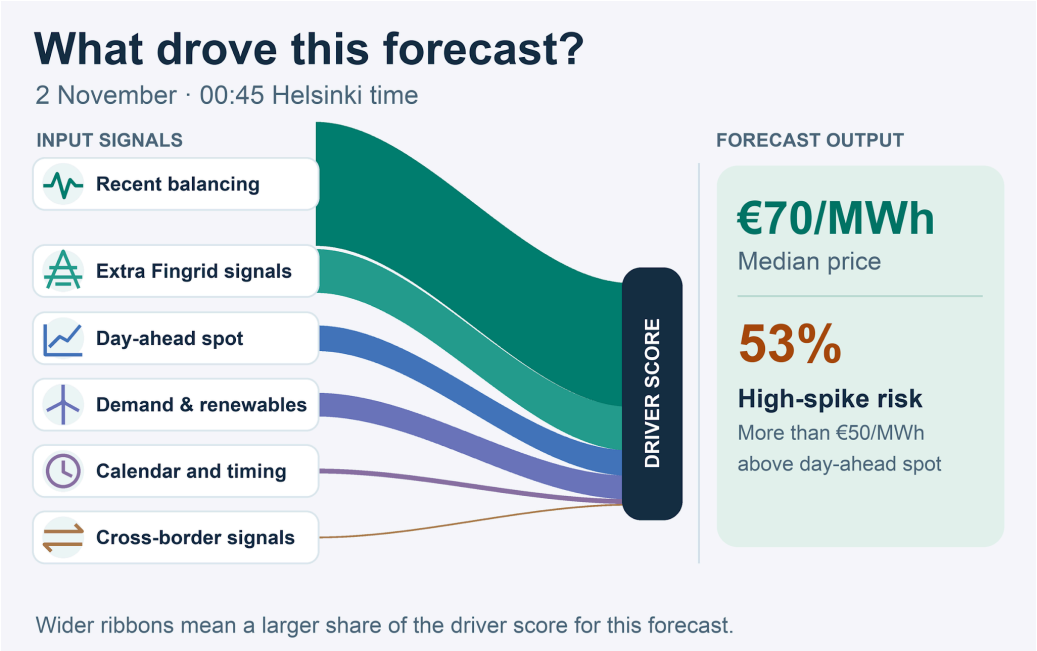



*Figure 3. The score combines contributions from the direct median and balancing-direction models. The price and spike risk at right are separate outputs of the full forecast.*

Both horizons use the starter-pack inputs below. The six additional Fingrid feeds are used only for the first two hours.

| Data group | What the model uses | Source |
|---|---|---|
| **Day-ahead price** | 1. Finnish day-ahead spot price<br>2. Shape of the spot curve | Nord Pool |
| **Demand and renewables** | 1. Consumption forecast<br>2. Wind forecast<br>3. Solar forecast | Fingrid |
| **Recent balancing** | 1. Recent imbalance prices and spreads<br>2. Dominant balancing direction<br>3. Up- and down-regulation reserve prices | Fingrid |
| **Cross-border conditions** | 1. Finland–SE1 power flow<br>2. SE1 and SE3 imbalance prices<br>3. Finland–Estonia transfer capacity | Fingrid and ENTSO-E |

| | | |
|---|---|---|
| **Calendar and timing** | 1. Hour and weekday<br>2. Holidays and daylight saving<br>3. Forecast lead time | Starter calendar and forecast timestamps |
| **Calculated grid imbalance** *(outside starter pack)* | 1. Calculated grid imbalance before activated reserves<br>*Published with a 20-minute delay.* | Fingrid Open Data · 397 |
| **Other near-term grid signals** *(outside starter pack)* | 1. Grid frequency<br>2. Surplus or deficit<br>3. Measured wind and consumption<br>4. Net electricity imports<br>*Updated every 3 minutes.* | Fingrid Open Data · 177, 186, 181, 193, 194 |

Starter-pack rows enter the forecast only at or after their recorded first_usable_origin_utc. For additional feeds, the model waits 30 minutes after each measurement period ends.

## 5 Validation results

| Validation metric | Organizer baseline | LSTGBM |
|---|---|---|
| Median price MAE (€/MWh) ↓ | 66.20 | **41.57** |
| Mean P10/P90 pinball loss ↓ | 24.18 | **11.77** |
| Mean spike Brier loss ↓ | 0.11318 | **0.09050** |
| Direction Brier loss ↓ | 0.45091 | **0.39297** |
| P10–P90 interval coverage (target 80%) | 68.3% | **81.3%** |

*Figure 4. Validation results, 7 September–4 October 2026 (26,700 scored forecast rows per metric). ↓ indicates that lower values are better.*

LSTGBM outperformed Turku Energia's baseline on all four scored metrics and came closer to the 80% interval coverage target.

## Appendix A. Feature ablation by forecast horizon

We removed one input group at a time and retrained the model. The table shows how much forecast error changed versus the full model, averaged over two 28-day validation folds.

| Inputs removed | Forecast error | Steps 1–8 | Steps 9–20 | Steps 21–40 |
|---|---|---|---|---|
| **Real-time Fingrid** | Balancing direction | **+9.7%*** | — | — |
| | Price spikes | **+5.4%*** | — | — |
| | Prediction interval | **+4.0%*** | — | — |
| | Median price | **+3.5%*** | — | — |
| **Lagged price and regulation** | Median price | **+4.1%*** | **+1.6%*** | +0.8% |
| | Balancing direction | **+4.3%*** | +2.6% | +1.0% |
| **Spot-derived** | Prediction interval | **+8.0%*** | **+9.2%*** | **+10.6%*** |
| | Price spikes | +7.7% | +7.7% | **+8.3%*** |
| | Median price | +0.3% | +0.2% | +0.6% |

*Figure 5. Change in forecast error after removing each input group, averaged over two 28-day validation folds. Asterisks mark statistically significant increases in both folds.*

Recent system data mattered most in the first two hours, especially for predicting balancing direction. Spot-derived inputs helped the model estimate how uncertain the price forecast was throughout the ten hours.

Removing any one of the other four input groups showed no statistically significant effect across both folds. These were day-ahead forecasts and transfer capacities, calendar variables, neighbouring-area price gaps, and flows and consumption errors.

## Appendix B. Model architecture ablations

These experiments tested three design choices across two 28-day validation folds.

| Design choice | Comparison | Validation result |
|---|---|---|
| **Direction-first mixture** | Same near/far models, with and without direction-conditioned price forecasts. | Adding the mixture lowered median-price MAE by **1.7%** and **2.3%**. |
| **Two-hour specialist** | One shared model versus a near/far split with real-time Fingrid inputs in the first eight steps. | For steps 1–8, direction Brier fell from **0.411 to 0.357** and from **0.397 to 0.345**. |
| **Interval widening** | The same mixture forecasts, with and without widening the prediction interval by 1.25×. | Without widening, the nominal 80% interval covered **74.5%** and **70.7%** of outcomes. |

*Figure 6. Validation results for three LSTGBM design choices. Paired results are shown in fold order.*

The specialist comparison measures the combined effect of splitting the model by horizon and adding real-time Fingrid inputs.